

Quiz 3

ECE 2201 – Spring 2025

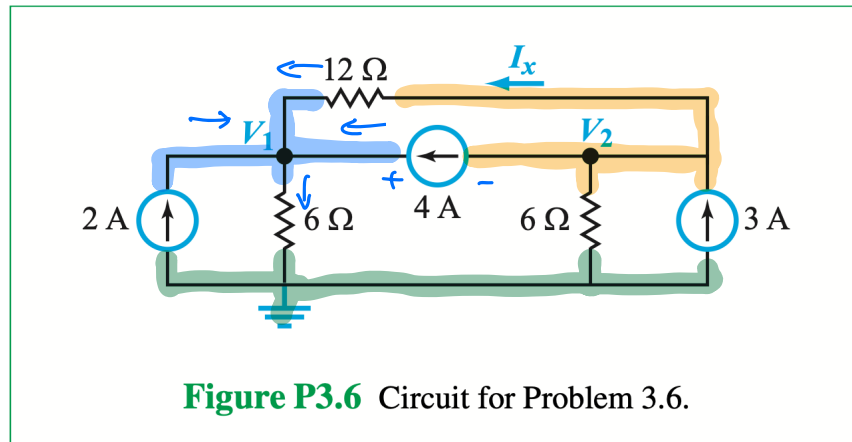
Instructions

- Do not open the quiz until the instructor announces the start
- You will have 15 minutes to complete the quiz
- Closed book and notes – no formula sheets
- You may use your calculator
- Annotate your units with square brackets → for example, [A] for Amps
- Write your answer in the space provided

Name: _____ PS ID: _____

For this problem, write the raw equations and the simplified equations in the spaces below for V_1 and V_2 .

3.6 Use the nodal-analysis method to find V_1 and V_2 in the circuit of **Fig. P3.6**, and then apply that to determine I_x .



$$\begin{aligned}
 & \times 12[\Omega] \left(2[A] + 4[A] + \frac{V_2 - V_1}{12[\Omega]} = \frac{V_1}{6[\Omega]} \right) & \left(3[A] = 4[A] + \frac{V_2}{6[\Omega]} + \frac{V_2 - V_1}{12[\Omega]} \right) \times 12[\Omega] \\
 & 72[V] + V_2 - V_1 = 2V_1 & 36[V] = 48[V] + 2V_2 + V_2 - V_1 \\
 & 3V_1 - V_2 = 72[V] & 3V_2 - V_1 = -12[V] \\
 & \swarrow & \nearrow \\
 & 3(3V_2 + 12[V]) - V_2 = 72[V] & V_1 = 3V_2 + 12[V] \\
 & 8V_2 + 36[V] = 72[V] & \hookrightarrow V_1 = 3(4.5[V]) + 12[V] \\
 & 8V_2 = 36[V] \Rightarrow V_2 = \underline{\underline{4.5[V]}} & = \underline{\underline{25.5[V]}}
 \end{aligned}$$

Raw equations	$2[A] + 4[A] + \frac{V_2 - V_1}{12[\Omega]} = \frac{V_1}{6[\Omega]}$ $3[A] = 4[A] + \frac{V_2}{6[\Omega]} + \frac{V_2 - V_1}{12[\Omega]}$
Simplified equations	$3V_1 - V_2 = 72[V]$ $3V_2 - V_1 = -12[V]$
I_x	$-1.75[A]$

$$I_x = \frac{V_2 - V_1}{12[\Omega]} = \frac{4.5[V] - 25.5[V]}{12[\Omega]} = -1.75[A]$$